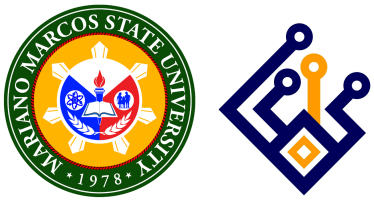
Adapto: An Adaptive Virtual Reality Learning Platform for Automata Theory and Formal Languages using Knowledge Tracing Algorithms

THESIS

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The Researchers

ABSTRACT

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Automata Theory and Formal Languages is a fundamental yet difficult subject for Computer Science students that often leads to high failure rates due to its abstract concepts and static traditional teaching methods. To address this, this study aims to develop Adapto: An Adaptive Virtual Reality Learning Platform for Automata Theory and Formal Languages using Knowledge Tracing Algorithms.

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1 Introduction

This chapter presents the background of the study, statement of the problem, objectives, significance of the study, scope and delimitation, and definition of terms.

1.1 Background of the Study

Automata Theory and Formal Languages form one of the most fundamental branches of Computer Science, underpinning compiler design, natural languages processing, digital circuit design, and the theoretical limits of computation itself [1].

Despite its foundation importance, the subject is widely regarded by students as one of the most abstract and difficult topics in the Computer Science curriculum. It replaces tangible, physical computers with imaginary machines, deterministic and non-deterministic finite automata, pushdown automata, and turing machines and requires learners to reason through heavy mathematical proof rather than observable, hands-on computation [1]. Students are commonly expected to grasp state transitions, grammar derivations, and language equivalence proofs through static diagrams and textbook notation alone, with little opportunity to manipulate, visualize, or experiment with these abstract constructs directly.

Traditional instructional methods compound this difficulty, lecture-based delivery paired with flat, two-dimensional diagrams limits how well students can visualize dynamic processes synch as state transitions during string acceptance or the step by step derivation of a string from a context-free grammar. Studies on visualization-based learning support this concern, finding that students perform significantly better with dynamic visualizations than with static representations [2]. Furthermore, conventional classroom instruction typically delivers uniform content to every learner regardless of individual pacing, prior knowledge, or specific misconceptions, leaving faster learners under challenged and struggling learners without the targeted remediation they need.

This difficulty is not merely anecdotal: empirical investigation into theory of computation courses has found that they consistently suffer from high failure rates, with only a minority of students performing well despite the courses' recognized importance in computer science curricula. Similarly, studies applying constructivist approaches such as Problem-Based Learning to Computer Theory instruction have noted the subject's high level abstraction and complexity as a persistent barrier to student motivation and learning, even when active-learning strategies are introduced.

Recent advances in immersive technology and adaptive learning offer a promising response to both limitations. Virtual Reality (VR) has been shown to improve the learning experience by turning traditionally passive or abstract lessons into immersive, interactive experiences [3]. Applied to Automata Theory, VR allows a learner to construct automata, add and remove states, draw transitions, and observe how a machine processes an input string step by step, turning an abstract proof into a spatial, manipulable experience. Related work has also explored automata theory through a gamification lens, finding that finite state machines are themselves a natural representation for gamified instructional media [4]. On its own, however, immersive visualization only addresses how content is presented; it does not resolve the deeper pedagogical problem or whether the right time. This is the role of Knowledge Tracing, a family of computational models that continuously infer a learner's underlying mastery state from their interactions and use that interface to personalize subsequent content, pacing and remediation.

This study is anchored on this second, more fundamental problem. While immersive removes the barrier of abstraction, it is the Knowledge Tracing Algorithms, not the VR medium itself, that determine whether instructions are genuinely responsive to each learner's evolving understanding. Therefore, we purpose Adapto, a VR-based learning platform for Automata Theory and Formal Languages, whose core contribution is a set of knowledge tracing algorithms: Bayesian Knowledge Tracing (BKT), its underlying Hidden Markov Model (HMM) formulation, and prospectively, Knowledge Space Theory (KST). These algorithms allow the system to infer a learner's probability of

mastery for each discrete skill after every interaction, and to use that inference to determine what the learner should see, review, or attempt next.

1.2 Statement of the Problem

This study focuses on determining how a Knowledge Tracing Algorithm can be integrated into a prototype VR-based learning platform to help Computer Science students better understand Automata Theory and Formal Languages. The abstract nature of the subject makes it difficult for students to grasp, and conventional means of instruction do not sufficiently accommodate the differences in students' prior knowledge and pacing.

Specifically, the study seeks to address the specific learning challenges students face in Automata Theory and Formal Languages, the profile of student-respondents, and the performance data needed to determine the teaching approach.

It also seeks to determine which Knowledge Tracing Algorithm is suitable for tracking students' mastery in terms of predictive accuracy, real-time computational performance, and suitability for tracking mastery of the topic's concepts.

Furthermore, it seeks to determine how the selected Knowledge Tracing Algorithm can be integrated into an immersive VR platform to dynamically adjust learning paths based on the students' knowledge states.

It also seeks to determine how accurately the integrated Knowledge Tracing Algorithm predicts and tracks individual students' progress and mastery.

Lastly, it seeks to determine the extent to which users accept the resulting platform in terms of functional suitability, usability, performance efficiency, reliability, and compatibility.

1.3 Objectives of the Study

This study aims to develop Adapto, an adaptive virtual reality learning platform that enhances students' understanding of and engagement with Automata Theory and Formal Languages through Knowledge Tracing Algorithms.

Specifically, it seeks to

1. Identify the learning challenges faced by students in Automata Theory and Formal Languages, decompose the pilot topic into trackables skills, profile

student-respondents, and gather the performance data needed to determine the teaching approach and inputs required for Adapto.

2. Determine the most suitable Knowledge Tracing Algorithm based on predictive accuracy, real-time computational performance, and suitability for tracking mastery of the topic's concepts.
3. Design and develop an immersive VR platform that integrates the selected Knowledge Tracing Algorithm to dynamically adjust learning paths based on students' knowledge states.
4. Assess the predictive accuracy and performance of the integrated Knowledge Tracing Algorithm in tracking individual students' progress.
5. Evaluate the level of user acceptance of Adapto in terms of functional suitability, usability, performance efficiency, reliability, and compatibility.

1.4 Significance of the Study

This findings of this study aims to directly benefit the following:

Instructors. The study will provide educators with an innovative, supplemental teaching tool designed to demonstrate the abstract and complex Automata concepts. With the use of an immersive VR environment, they can better visualize theoretical frameworks during classroom discussions, increase student engagement, and deliver complex topics through structured and interactive visual models.

Students. Adapto may provide students with an interactive VR environment that simplifies the visualization of Automata concepts that will serve to them as an adaptive learning tool. With the use of knowledge tracing and adaptive algorithms, the system dynamically adjusts the task difficulty, content delivery, and pacing based on individual student performance metrics and learning speed. This personalized approach improves active participation, strengthens problem-solving capabilities, and improves the overall mastery of the course.

Future Researchers. This study may serve as a foundational baseline for future research focused on VR-based educational applications, adaptive systems, and Knowledge Tracing Algorithms. This study is intended to surface early findings, feasibility considerations and comparative insights among the evaluated algorithms in this study that may help future

researchers extend the system to the remaining topics in Automata Theory and Formal Languages, and to larger-scale deployment.

1.5 Scope and Delimitation of the Study

This study will focus on the development of a virtual reality-based learning platform integrated with a Knowledge Tracing Algorithm to enhance the understanding and engagement of Computer Science students in Automata Theory and Formal Languages. It will be designed to provide an interactive and immersive experience through which students can visualize key concepts, including Set Theory, Deterministic Finite Automata (DFA), Non-deterministic Finite Automata (NFA), Regular Expressions, State Elimination, Context-Free Grammars (CFG), Derivation Trees, Pushdown Automata, and Turing Machines. Due to the scope and technical demands of VR development, the system will be limited to these identified topics and will not extend to other areas of theoretical computer science. The platform will integrate an adaptive learning path that adjusts in accordance to the learner's assessed knowledge state, as well as a knowledge tracing component responsible for monitoring and predicting individual student mastery over time. The development of this study will be carried out using the Meta Quest 2, Blender, and Godot Engine.

This study will be conducted among Computer Science students of the College of Computing and Information Sciences, and will be tested using available VR tools, such as the Meta Quest 2, within the institution.

1.6 Definition of Terms

To provide a better understanding of the study, following terms are defined below.

Adapto refers to the adaptive virtual reality learning platform developed in this study for teaching Automata Theory and Formal Languages using Knowledge Tracing Algorithms.

Adaptive Learning refers to an educational approach in which learning content, activities, pacing, or difficulty are adjusted based on the learner's estimated knowledge, performance, and learning needs.

Automata Theory refers to the study of abstract computational machines and the mathematical models used to describe computation and formal languages.

Knowledge Tracing refers to the computational process of estimating a learner's evolving knowledge or mastery of specific skills from their sequence of learning interactions and responses.

Knowledge Tracing Algorithms refer to computational models used to estimate and update a learner's knowledge state based on their interaction history.

Learner Knowledge State refers to the estimated level of a learner's mastery of a particular skill at a given point during the learning process.

Skill refers to a specific, assessable component of knowledge within a learning topic.

Diagnostic Assessment refers to an assessment administered before instruction to determine a learner's existing knowledge and identify areas of strength and weakness.

Learner Interaction refers to an observable action or response produced by a learner while using the system, such as answering a question, completing an activity, or interacting with a learning task. These interactions provide the data used by the Knowledge Tracing Algorithms.

Virtual Reality (VR) refers to a computer-generated interactive environment that allows users to experience and interact with simulated objects and spaces. In this study, VR provides the immersive environment through which learners interact with Automata Theory learning activities.

Prototype refers to an early, limited-scope working version of Adapto developed to demonstrate the technical and pedagogical feasibility of integrating a Knowledge Tracing Algorithm into an immersive VR platform, as distinguished from a complete, production-ready system.

Pilot Deployment refers to the limited-scale administration of the Adapto prototype to a small group of student-respondents, used to gather performance data, assess predictive accuracy, and evaluate user acceptance within the scope of this study.

Student Performance Data refers to the structured dataset of learner responses — including correctness, timestamp, and associated skill — collected during pretest, in-VR interaction, and posttest, used as input to the Knowledge Tracing algorithms.

2 Review of Related Literature

This chapter presents related concepts, literature, and systems reviewed by the researchers. These will be used as support for identifying the features for the development of Adapto.

2.1 Related Concepts

2.1.1 Challenges in Automata Theory and Formal Languages

Automata Theory and Formal Languages is one of the fundamental branches of theoretical Computer Science that explores core mathematical frameworks used to model computation, formal grammars, compiler design, and digital logic.

Automata Theory and Formal Languages is generally considered as a mandatory course for the undergraduate computer science program. The curriculum introduces students to several areas including familiarizing with the foundations and principles of computer science and strengthening students' ability to carry out formal and rigorous mathematical arguments [5]. The core topics includes Set Theory, Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NFA) for regular languages, Regular Expressions, State Elimination, Context-Free Grammars (CFG), Derivation Trees, Pushdown Automata, and Turing Machines [5]. Mastery of these concepts is important for computer science education as it helps students build good problem-solving skills and logical thinking for advanced software engineering, syntax analysis, and language processing [5], [6].

Despite its academic significance, Automata Theory has been considered as one of the most challenging and abstract subjects for students taking the undergraduate degree program, resulting in high failure rates in their classes [7] . These challenges primarily stem from the abstract nature of the theoretical concepts introduced, mathematical concepts, the mental

visualization of complex state transitions, high cognitive load, and fluctuating academic motivation [6], [7]. Unlike other courses in computer science that provide immediate execution feedback and visual outputs, Automata has traditionally been taught using static and text-heavy methods, which makes the dynamic and interactive nature of the field hard to visualize [6].

Recent studies validate these struggles, demonstrating that learning barriers stem directly from the subject's complexity, traditional teaching methods, and diverse student backgrounds. Specifically, students find it hard to grasp static computational models, face confusion when shifting from deterministic to non-deterministic models, and make procedural errors when translating grammars [6], [7].

To address these challenges, educational research emphasizes the necessity for digital technologies and adaptive educational interventions as it helps educators provide students with a more engaging and effective learning experience [6]. Because traditional teaching methods are often ineffective for managing different student needs in abstract courses, effective modern frameworks require systems capable of profiling student-respondents, breaking down complex pilot topics into granular, trackable micro-skills, and systematically gathering baseline performance data [6], [7]. With the help of these adaptive tools, it can dynamically help and improve their teaching approaches and instructional inputs, delivering targeted remediation and personalized feedback to guide students toward successful mastery [6], [7].

2.1.2 Knowledge Tracing Algorithms

While gathering student performance data and breaking down pilot topics into trackable skills help in establishing input requirements for an adaptive tool, the system still needs a computational mechanism to interpret learner interactions over time.

Knowledge tracing (KT) refers to the task of modelling a student's knowledge over time. It aims to observe, represent, and quantify a student's knowledge or skill such as the mastery level of the skills based on a trace of interactions with the learning activities and materials [8], [9]. Over time, different and new knowledge tracing algorithms exist and are continuously being proposed and modified. These KTs are often used to provide personalized tutoring to students through intelligent tutoring systems [9]. Selecting an optimal knowledge tracing algorithmic approach requires thorough evaluation including its predictive accuracy, real-time computational performance, and suitability for tracking mastery of the conceptual topics.

Bayesian Knowledge Tracing (BKT) has been considered as the most prominent probabilistic graphical model for knowledge tracing [9]. Throughout the learning journey of a learner, BKT offers a dynamic process of modeling the probability that processes knowledge of each skill. Based on their interactions with educational tasks or assessments, its goal is to estimate the knowledge state of individual learners over time [10]. BKT assumes that knowledge is binary that indicates whether the learner has learned the skill and suggests that it cannot be unlearned or forgotten [9], [10]. While this structural simplicity makes BKT interpretable, its assumption of permanent retention can limit its suitability for tracking mastery in complex conceptual domains where forgetting curves or skill degradation occur.

Closely related to probabilistic modelling, Hidden Markov Model (HMM) is one of the most widely used statistical models for sequential data analysis [11]. HMM is a stochastic dynamical system in discrete time whose state is considered "hidden" or not directly observable, accompanied by a sequence of noisy observations that are functions of the hidden state and each point in time [11]. In educational data mining, HMMs generalize the principles underlying sequential tracking by allowing more flexible transition and emission probabilities between

latent cognitive states and observable student responses. However, applying HMMs to real-time adaptive systems requires careful consideration of their computational complexity, particularly when handling long interaction sequences or high-dimensional skill spaces. Parameter estimation via Expectation-Maximization (such as the Baum-Welch algorithm) scales quadratically with the number of hidden states, making standard HMMs less ideal for fast, real-time feedback compared to similar models [9], [11].

To address the limitations of treating skills as completely isolated dynamic processes, Knowledge Space Theory (KST) shifts the paradigm from purely probabilistic temporal modeling to a structural, domain-oriented evaluation framework [12]. Developed in mathematical psychology, KST is a set-theoretical framework that assesses individual knowledge states and guides learning paths [12]. Rather than evaluating isolated skills independently, KST represents a knowledge domain as a structured set of problems or concepts constrained by prerequisite dependencies. While classical KST relied on binary assessment, recent advances in polytomous knowledge spaces extend this structure to capture multi-level mastery and partial understanding, utilizing atomic state representations to enhance computational efficiency during state extraction and domain lookup [13]. Under KST, a learner's knowledge state is defined as the exact subset of concepts they are capable of solving at a given time, allowing the system to pinpoint prerequisite gaps when a learner struggles [12], [13].

Comparing BKT, HMM, and KST across the criteria of predictive accuracy, real-time computational performance, and suitability for concept mastery shows key structural trade-offs across all three approaches [8]-[13]. In terms of predictive accuracy, BKT offers a stable baseline but plateaus due to binary state assumptions [9], [10], whereas HMM improves accuracy through multi-state latent progressions [11]. KST provides high structural accuracy by evaluating

prerequisite competencies, with modern polytomous extensions capturing partial understanding and multi-level scoring [12], [13]. Regarding real-time computational performance, BKT achieves optimal $O(1)$ efficiency per interaction [9], while standard HMMs suffer from $O(N^2 \cdot T)$ quadratic limitation during extended sequences [9], [11]. KST balances both by utilizing fast runtime subset lookups, with atomic representation algorithms minimizing state extraction overhead during live sessions [12], [13]. Finally, for tracking concept mastery, BKT and HMM evaluate skills independently or sequentially, failing to represent cross-topic dependencies [9]-[11]. In contrast, KST natively incorporates prerequisite dependency networks to diagnose foundational gaps and structurally map topic mastery in adaptive environments [12], [13].

2.1.3 Knowledge Tracing in Adaptive Virtual Reality Learning

Integrating Knowledge Tracing (KT) into Virtual Reality (VR) environments enables the dynamic modeling of learner mastery to deliver better instructions and personalize feedback [14]. While traditional teaching methods rely heavily on questionnaires and pre- and post-tests methods; immersive environments expand KT capabilities by capturing learner's continuous multimodal interactions, such as eye-movements, head pose, and physical interaction trajectories [14], [15].

Beyond just observational tracing on the learner, using KT algorithms within immersive frameworks improves the design and development of fully adaptive VR platforms [15]. In these platforms, the KT algorithm continuously computes the students' evolving latent knowledge state by processing their continuous performance and multimodal behavioral metrics [15], [16].

Advanced KT models allow the system to accurately keep track of what the student has mastered or understood, even during complex, multi-step activities [16].

Instead of presenting a static, linear series of virtual scenarios, the VR system uses these estimated knowledge states to instantly adapt and adjust learning paths in real-time [15], [17]. For example, if a learner demonstrates high mastery or rapid task efficiency, it automatically skips easy steps and moves them to more challenging tasks [17]. On the other hand, if a learner makes repeated mistakes or gets confused, the platform automatically provides helpful hints, extra guidance, or simpler practice tasks [15], [17]. By integrating the KT models directly into the VR experience, adaptive learning platforms keep students engaged, prevent learner frustration through continuous, personalized instructions to help them learn more effectively [15], [17].

2.1.4 Knowledge Tracing Performance Evaluation

The effectiveness of KT algorithms is measured mostly by the capability to estimate the state of knowledge dynamically and make accurate predictions about future student responses [8]. Standard evaluation methodologies partition student interaction sequences into training and testing subsets using an 80:20 split to rigorously measure generalization performance [8]. In order to measure the accuracy of predictions on educational data sets, two metrics should be used: the Area Under the ROC Curve (AUC) and classification accuracy [18]. In contrast to the latter which is defined as the ratio of correct binary predictions at a particular decision threshold, AUC is a probabilistic metric that calculates the accuracy of predictions of the KT algorithm at all possible thresholds [18].

Various algorithmic architectures show their own strengths during student behavior prediction [19]. Classical probabilistic algorithms such as Bayesian Knowledge Tracing (BKT), estimate the degree of skill mastery separately but take into account human cognitive processes like slipping and guessing [19]. Modern deep learning algorithms including Recurrent Neutral Network (RNN) and attention-based architecture can trace joint dependencies and dynamic interactions among several skills [19]. According to recent comparative studies, neural and attention-based algorithms demonstrate better accuracy and high AUC scores on benchmarks, and classical and simple architectures have their superiority in terms of computational effectiveness and results interpretation [19].

Moreover, for tracking individual learning trajectory, one should not neglect the importance of the initialization process, which defines the estimation of parameters based on student interactions at the beginning of the learning process such as pre-tests or quizzes [20]. Recent research suggests that the optimization of the duration and number of interactions in the initialization period helps to enhance the performance of KT algorithms by providing sufficient history and constraining parameters of student proficiency [20].

2.1.5 Adaptive and Personalized Learning in Higher Education Context

Adaptive and personalized uses technology to adjust learning content, pacing, and assessment to each learner's performance and needs, with the aim of improving learning outcomes and engagement [21], [22]. Higher education institutions continue to struggle with student success and retention, and standardized, one-size-fits-all instruction does little to address students' varied backgrounds and prior knowledge [22], [24], [25]. For this reason, a growing number of studies have examined how effective adaptive learning is in university settings, and recent reviews now provide a reasonably clear picture of what it can be cannot deliver [22], [24].

The strongest evidence concerns academic performance. In a scoping review of 69 studies involving undergraduate students, 41(59%) reported improved academic performance, with gains seen in post-test scores, exam marks, and critical thinking skills, and some studies reported significant improvement among low-achieving students [22]. A systematic review of AI-driven adaptive systems in higher education likewise reported consistent positive effects in performance and engagement across its 15 included studies [24]. In a controlled experiment across university courses, students who used an adaptive platform obtained higher mean grades than a conventionally taught control group (82.5 versus 76.3,$p < 0.001$), and more time and interaction on the platform were associated with higher grades [25]. This association is correlational and should not be read as proof of cause. Effectiveness is not universal, however: 28 of the 69 studies in [22] found no effect on academic performance, and an experimental study of adaptive courseware found that the benefit went mainly to better-performing students [26].

Adaptive learning has also been reported to influence engagement and the learning process itself. Twenty-five of the 69 studies in [22] (36%) reported increased student engagement, and in [25] students valued the personalized content and immediate feedback, while educators observed better understanding and retention of complex concepts. Some studies also reported gains beyond grades, such as improved self-regulation strategies [22]. Mixed results exist here as well: one study in [22] reported reduced engagement because individual learning pathways left students isolated, and another found that students' self-regulated learning scores decreased after months on an adaptive platform. Intelligent assistants that deliver personalized quizzes and learning pathways are likewise designed to reduce learners' cognitive load, although evidence for this comes mainly from system design rather than controlled outcome studies [23].

The reviewed literature suggests that effectiveness depends heavily on the subject and on how the system is implemented. Results are more consistent in STEM fields, where content can be broken into measurable steps and learner performance is easier to track [22], [24]. In [25], the platform's effect was significant in mathematics and science but not in humanities and languages. Reviews also caution that most evidence covers short-term outcomes, that implementation depends on infrastructure and faculty readiness, and that the instructor's role remains central [22], [24]. For Adapto, this evidence indicates that an adaptive approach is well suited to a structured, skill-based topic such as Automata Theory, and that pre-knowledge assessments, the most common adaptation trigger in the reviewed studies [22], are an appropriate basis for adapting content. Because the evidence for the effectiveness is mixed, Adapto's effectiveness is evaluated directly through posttest result and live-session predictive accuracy rather than assumed.

2.1.6 Game-Based Learning in Higher Education Context

Game-Based Learning (GBL) is an instructional approach in which games are used to foster and support the process of learning and skill development using problem-solving, challenges, engaging activities and meaningful feedback. Game-Based Learning grew when the education sector was also affected by the Paradigm shifts to a more online approach [27]. In Higher Education, GBL gained attention as an approach to improve student engagement while providing opportunities to apply theoretical concepts through interactive learning experiences [28]. GBL was found to be associated with improvements in motivation, reflective thinking, creative thinking, problem-solving, computational thinking, and student engagement. These findings suggest that game-based activities can support learning beyond the memorization of

concepts by encouraging students to actively participate in tasks and apply their understanding [28].

Despite its potential benefits, the implementation of GBL in higher education also presents challenges [29]. In a systematic review of higher education educators perspectives, identified improved engagement, student interaction, and collaborative learning as factors encouraging the adoption of GBL and gamification. However, limited development time and uncertainty about its educational benefits were some barriers reported. These findings emphasize that educational games should be designed with clear learning objectives and evaluated for their educational and technical effectiveness rather than relying solely on their interactive features [29].

Ultimately, At the core of game-based learning is the fact that the learners can learn from their mistakes as they explore multiple different solutions on each problem [27]. However, common misconceptions about the adoption of GBL is that you can solely use it to learn, but studies have stated that it is merely a tool to supplement the actual curriculum and does not in fact replace the curriculum [27], [28]. Ultimately, by tapping into the students' interests and providing active and interactive learning experiences, educators can establish a positive learning environment that fosters students' learning abilities and enthusiasm [28].

2.2 Related Systems

2.2.1 Automata Factory

Automata Factory is a single-player 3D educational game developed in Unity for teaching Automata Theory and Formal Languages to college students. The game employs a low-poly factory environment with text-based interaction [24]. The game's content encompasses Regular Expressions, Deterministic Finite Automata (DFA), and Non-deterministic Finite Automata (NFA), organized as separate topics, each having a set of levels preceded by an introductory topic. The interface displays the learner's progress for each topic as a completion count.

The game demonstrates that automata concepts may be embedded in a themed game environment and organized into a structured sequence rather than being presented solely as static diagrams. However, its progress tracking records completion rather than the learner's actual level of understanding [24].

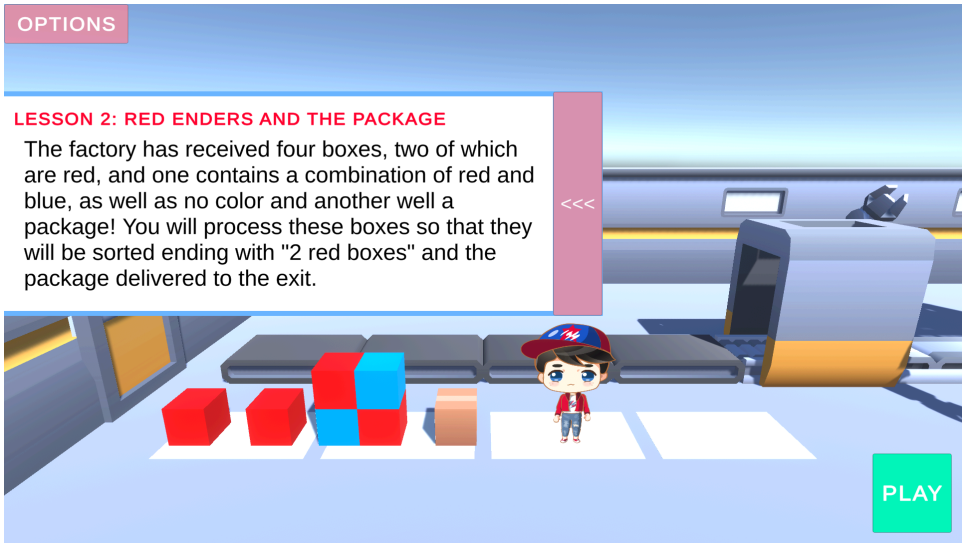


Figure 1: Screenshot of Automata Factory

2.2.2 finsm.io

finsm.io is an open-source, browser-based application for teaching finite automata that enables the user to create, simulate, and export Deterministic and Non-deterministic Finite Automata, and has been used in finite automata courses across years. It automatically detects the alphabet of a language, and identifies errors in constructed machines [25].

The tool illustrates the educational value of allowing learners to construct and test their own machines rather than merely observing completed examples. However, its coverage is restricted to finite automata, and it also does not provide guided instructions nor tracks learner mastery [25].

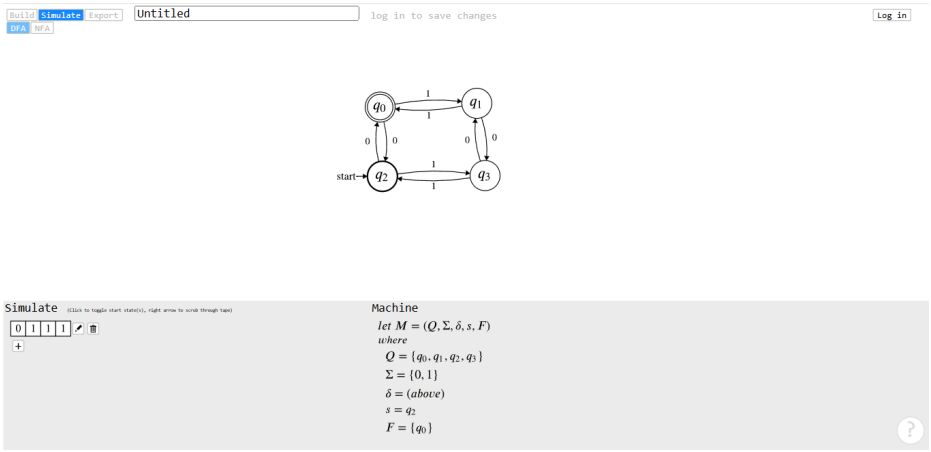


Figure 2: Screenshot of finsm.io

2.2.3 AutomataTutor 2.0

AutomataTutor 2.0 is a stand mobile application that extends the original AutomataTutor tutorial-and-sandbox app with a game-based competition system. Students take part in timed competitions, usually held during a lecture slot, where they solve a sequence of automata puzzles in order, and each correct solution unlocks the next challenge. In the automaton puzzle editor, the states and transitions of a reference automaton are scattered as separate pieces across a 6x6 grid.

The learner drags, rotates, and modifies these pieces to rebuild an automaton that matches a given regular language, and the system automatically verifies the result. A second editor, designed like a calculator, lets learners write regular expressions using context-relevant alphabet symbols and operators. It includes syntax highlighting and validation, and it gives counterexamples when an expression is logically incorrect, completion time and number of attempts (up to 2,100 points per questions), and the scores feed a public leaderboard. Instructors can create questions with an enhanced version of the grid editor and monitor student performance through analytical dashboards [26].

The system's coverage is limited to regular languages. Its first competition covered finite automata construction (11 questions), and its second covered regular expression construction (12 questions). In a six-week study with 196 undergraduate students, those who joined the voluntary competitions showed higher median improvement between class tests than the overall cohort. The reported advantages were 4.19 percentage points after the first competition and 5.31 after the second. Survey respondents (n=23) also reported greater motivation and better perceived problem-solving skills [26]. The tool demonstrates that automata construction itself can be turned into a puzzle with immediate feedback and competitive elements. However, the authors report only correlation, with possible self-selection bias, and extending the system to pushdown automata and Turing machines is listed as future work. Its scoring also reflects speed and attempt count rather than a learner's actual mastery, and it does not adapt content to individual learners [26].

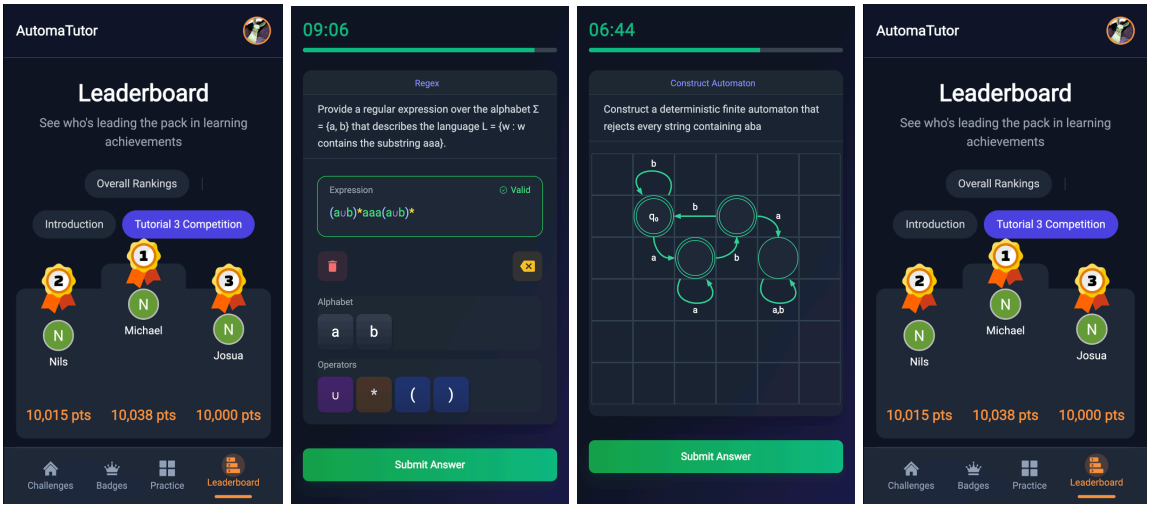


Figure 3: Screenshots of AutomataTutor 2.0

2.2.4 Immersive Virtual Environment for Data Structures and Algorithms

This system is an immersive learning environment developed in Unity that teaches Data Structures and Algorithms concepts like stacks and queues through quests of increasing difficulty accompanied by a progress tracker, and it is accessible in a web-based version and a virtual reality headset version [27]. A pretest and posttest administered to forty students using the web version showed an increase in the average score from 3.05 to 3.9 on a five-question test, while the performance on practical application questions remained unchanged. In the virtual reality version, eight participants rated its usability positively, although its immersion score fell below the baseline, with participants citing small buttons and limited interactive features [27].

The system demonstrates that an abstract computing topic can be taught through a structured, progress-tracked sequence of activities. However, the paper does not describe any adaptation of the quests to each learner's mastery.

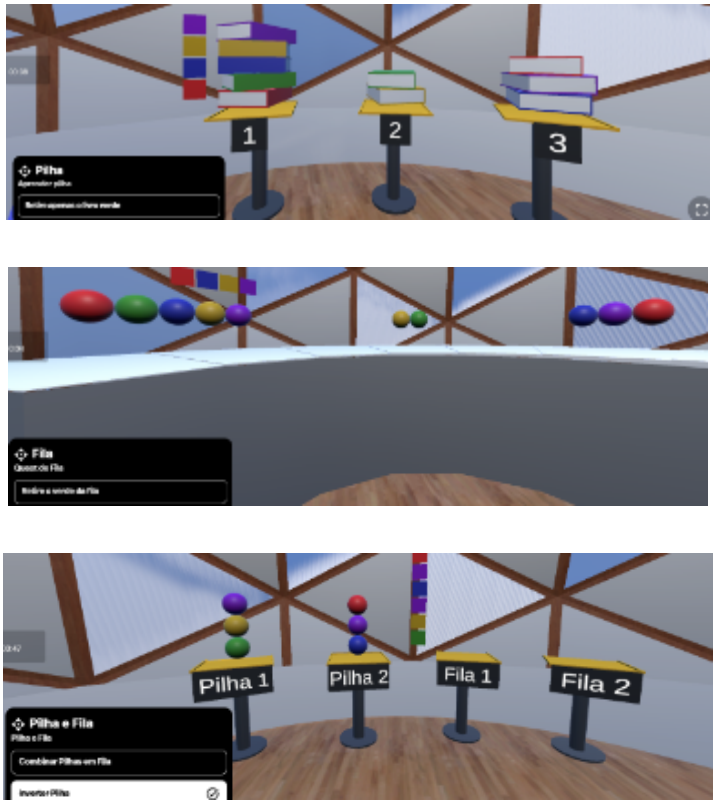


Figure 4: Screenshots of Immersive Virtual Environment for Data Structures and Algorithms

3 Methodology

This chapter presents the research design, the development framework used to build the system. The procedures followed in designing, implementing, and evaluation of the Algorithm models.

3.1 Research Design

This study follows an approach appropriate for studies whose primary output in a functioning artifact developed to address an identified problem, evaluated for both its technical performance and its practical utility [28]. The study additionally applies a comparative experimental design in evaluating the three candidates for Knowledge Tracing Algorithms: Bayesian Knowledge Tracing (BKT), Hidden Markov Model (HMM), and Knowledge Space Theory (KST).

3.2 Method

3.2.1 Data Gathering

Identify and gather the data needed for the Knowledge Tracing Algorithms to determine the Initial Data for the Algorithm Design:

1. Gathering profile by identifying student respondents needed for the Algorithms such as students who are proficient, intermediate or unknowledgeable about the subject matter.
2. Scoring, using a skill-tagged pretest and posttest are administered, each item mapped into one or more of the six DFA identified for this study: Simulation, Identification of Diagram, DFA Definition and Parts, DFA Building, DFA from Set Builder, and DFA from List.

3. Data Gathering using a simple test, The collected responses are examined to inform the initial parameters and values of Bayesian Knowledge Tracing (BKT), Hidden Markov Model (HMM), and Knowledge Space Theory (KST). For BKT, the assessment results inform the initial probability of knowing a skill and may contribute to estimates of guessing and slipping probabilities when supported by an appropriate method for identifying or estimating mastery.
4. Data cleaning processed the outputs from Data Gathering by arranging the results so each response is attributed to every skill assessed and correctness, to form per-skill sequential responses records. A manual review and judgement or borderline records before the dataset is passed to Algorithm Refinement.

3.2.2 Research Instrument

An 15-item questions, skill-tagged pretest/posttest is used to establish $P(L_0)/\pi$ and to measure learning gains per skill. The ISO/IEC 25010-based questionnaire gathers user acceptance data.

3.2.3 Algorithm Comparison and Testing

1. Collection of respondent profile data: Gender, GWA in Data Structures and Algorithms, and Year Level.
2. Administration of the pretest.
3. Distributing the respondent to three distinct test groups to observe algorithm results
4. Guided use of the VR Prototype, including its embedded test, producing the Result.
5. Data Cleaning and merging of Profile, Score, and Result into the structured dataset used for Algorithm Refinement.

6. Following full system integration administration of the posttest and live deployment session to validate predictive accuracy.
7. Administration of the ISO 25010-based acceptance questionnaire.

3.2.4 Data Pre-processing

Following the data collection phase, we construct a consolidated dataset comprising student profiles, pretest and posttest scores, and outcomes derived from the VR prototype. This dataset is subsequently prepared for analysis using three Knowledge Tracing Algorithms. All pre-processing procedures are conducted manually to ensure that records are carefully examined and resolved on a case-by-case basis. The process in implemented through the following steps:

1. **Record screening**. Incomplete records, such as respondents who did not finish the pretest, the VR session, or the posttest, are removed. Duplicate responses caused by double submission or repeated logging are also removed.

2. **Skill attribution**. Items tagged to more than one skill are expanded so that each response is counted for every skill the item assesses. This lets each skill keep its own responses history.

3. **Response encoding.** Each response is encoded, and sorted out if it is correct or wrong.

4. **Sequencing.** Responses are ordered by respondent and skill, following their timestamps, to form per-skill sequences of correct and incorrect responses. BKT and HMM use these sequences to update a learner's mastery estimate after each response. For KST, each respondent's responses across the six DFA skills are summarized into a single response pattern, which is matched to the nearest knowledge state.

5. **Profile linking**. Each respondent's profile (Gender, GWA in Data Structures and Algorithms, and Year Level) is attached to their records so that results can be described by the respondent group.

6. **Train-test split**. The cleaned dataset is divided once into a training set and a testing set (for example, 80% and 20%), and this same split is used for BKT, HMM, and KST so that their predictive accuracy is compared fairly. The split is made by students rather than by individual response. If one student's responses appeared in both sets, the test set would contain data the model had already seen, which would inflate its accuracy.

3.2.5 Selecting the Optimal Knowledge Tracing Algorithm

Select the optimal knowledge tracing algorithm, among Bayesian Knowledge Tracing (BKT), Knowledge Space Theory (KST), and Hidden Markov Model (HMM), based on predictive accuracy, real-time computational performance, and suitability for tracking mastery of the topic's concept. All three algorithms are run on the same preprocessed dataset and train-test split described in 3.2.2, and each is scored on three criteria. For predictive accuracy, each algorithm predicts whether a student will answer the next item correctly for every response in the testing set. BKT and HMM produce probabilities. KST predicts a skill as correct if it is inside the learner's knowledge state and incorrect if it is outside, so all three are also compared on classification accuracy using a 0.5 cutoff. For real-time computational performance, the average and longest time needed to update a learner's mastery after one response is measured on the hardware used to run Adapto, since updates must happen without a noticeable delay in the VR environment. For suitability for tracking mastery of the topic's concepts, we assess whether each algorithm

can represent the dependencies among the six DFA skills and whether its output can directly decide what Adapto shows next. An algorithm must first meet the real-time requirement, the one with the highest predictive accuracy among the remaining algorithms is preferred, and suitability decides between algorithms whose accuracy is close.

3.2.5.1 Bayesian Knowledge Tracing (BKT)

The theory of Bayesian Knowledge Tracing (BKT) estimates, for each skill separately, the probability that a learner has mastered it, and updates this estimate after every response [29]. It treats mastery as either known or not known. A correct response raises the estimate and an incorrect response lowers it, with each update adjusted for the chances of guessing and slipping and for the chance that the learner learned the skill during the attempt. Once P(L) for given skill reaches a pre-defined mastery threshold (e.g.,P(L) > 0.95), the skill is considered mastered For each skill, BKT tracks:

- **P(L_0):** Probability of already knowing the skill, estimated from pretest correctness.
- **P(T):** Probability of learning the skill after an opportunity, updated after each opportunity [30].
- **P(S):** Probability of slipping (answering incorrectly despite knowing), estimated from responses of learners considered knowledgeable.
- **P(G):** Probability of guessing correctly despite not knowing.

The observation and update equations are:

- Probability of a correct response:

$$P(correct)=P(L)(1-P(S))+(1-P(L))P(G)$$

- Probability of a wrong response:

$$P(wrong)=P(L)P(S)+(1-P(L))(1-P(G))$$

- Updated probability of knowing the skill if the response is correct:

$$P(Lnew)=P(L)(1-P(S))+(1-P(L))P(T)$$

- Updated probability of knowing the skill if the response is wrong:

$$P(Lnew)=P(Lwrong)+(1-P(Lwrong))P(T)$$

3.2.5.2 Knowledge Space Theory

Knowledge Space Theory represents what a learner knows as an exact set of mastered skills, called a knowledge state, and uses the prerequisite relations among skills to decide which states are possible [31]. For DFA, the prerequisite relations are set so that DFA Definition and Parts comes before Identification of Diagram and Simulation, Identification of Diagram comes before DFA from Set Builder and DFA from List, and DFA Building requires Simulation, DFA from Set Builder, and DFA from List. Under these relations, only 12 of the 64 possible combinations of the six skills are valid knowledge states, since a learner cannot realistically have mastered a skill without its prerequisites. To use it, each response from the pretest and the VR activities is matched to the skill it assesses, and the learner's overall response pattern is compared with the valid knowledge states. The state that best explains the pattern, meaning the one needing the fewest careless errors or lucky guesses, is taken as the learner's current knowledge state and is updated as new responses come in. The output is this knowledge state (the set of skills the learner has mastered) together with the outer fringe, which is the skills not yet mastered whose prerequisites are all in the state. Adapto presents activities from the outer fringe next and holds back skills whose prerequisites are still missing. If the learner struggles with a skill, the missing prerequisite is the first thing reviewed. For comparison with BKT and

HMM, the knowledge state is converted into a mastered or not-mastered indicator per skill. The practical use of this approach in adaptive assessment has been tested in a randomized comparison against newer adaptive engines [32]

3.2.5.3 Hidden Markov Model

The mastery per skill is represented as one of three ordered hidden state:

- Beginner (0-25%)
- Developing (25-85%)
- Mastered (85-100%)

The model is defined by **Initial State Probabilities (π)**, derived per learner from pretest performance on that skill. The **Transition Matrix (A)**, is the probability of moving between mastery states after a learning opportunity, estimated from pilot data and the **Emission/Observation Matrix (B)**, the probability of a Correct/Wrong response given the learner's true state, estimated through in-person testing sessions. The learner's most likely current state is inferred via the standard forward algorithm [33], updated after each new observation.

3.2.6 Algorithm Refinement

The algorithm that performs the best in the comparison in 3.2.5 is the one we improve. Instead of creating a new method, we address its weaknesses identified in the comparison, such as skills or learners it predicted poorly or situations where updates were slow. We also review the unselected algorithms to identify strengths that can be integrated into the chosen one. For example, if BKT is selected, we might add more mastery levels like HMM or use KST's prerequisite order among the six DFA skills to decide the next skill to teach; if HMM is chosen, we may adopt BKT's simple mastery threshold; and if KST is selected, we might include probability-based updates from BKT and HMM to handle careless errors and lucky guesses. All changes are made and compared using the training set with a validation portion held out by student, keeping the testing set unused until the final evaluation. The edited algorithm is then tested once on the testing set using the same metrics as in Section 3.2.5, and it is kept only if it improves predictive accuracy while meeting real-time requirements. This improved algorithm is

integrated into Adapto as described in Section 3.2.7 and may be further refined with live data as outlined in Section 3.2.8.

3.2.7 Developing the Adapto

The VR prototype developed in Objective 1 serves as the foundation for the final application, with the algorithm selected in Objective 2 integrated into its environment, interactions, and skill-based activities. This integration is refined through Agile development cycles [34], where each iteration involves connecting learner interactions in real time to the algorithm, updating mastery estimates, and using those updates to adapt the content, activity difficulty, or remediation presented. The process emphasizes continuous testing and adjustment to ensure the adaptive logic works seamlessly within the VR environment. The expected outcome is a functional VR application in which learner actions directly influence mastery estimates, and those estimates visibly alter the learning path. If mastery updates fail to affect what the learner experiences next, the objective is considered unmet, even if the prototype or algorithm functions correctly in isolation.

3.2.8 Assess Predictive Accuracy of the Integrated Algorithm

Validation of the integrated Knowledge Tracing Algorithm focuses on its predictive accuracy once deployed inside the final VR application with real learners. Rather than relying on preliminary datasets as in Objective 2, this stage compares the algorithm's predicted mastery trajectory during live sessions against respondents' posttest results and observed skill mastery, using the same accuracy metrics (AUC/RMSE). The expected outcome is a quantified measure of predictive performance under real usage conditions, directly comparable to earlier benchmarks. If the live-deployment accuracy proves substantially lower, it indicates that the algorithm's parameters fail to generalize from development to authentic use, requiring methodological adjustments such as recalibration on live data.

3.2.9 Evaluate User Acceptance of Adapto

Evaluation of user acceptance for Adapto centers on four software quality characteristics: functional suitability, performance efficiency, reliability, usability, and compatibility; by using the ISO/IEC 25010 model [3]. After engaging with the VR application, respondents complete a Likert-scale questionnaire covering these dimensions, and results are analyzed through weighted means per characteristic and overall. The expected outcome is a quantified acceptance rating with descriptive interpretation. If one or more characteristics receive poor ratings despite the system functioning as intended, this points to usability or design issues rooted in Objective 3's design decisions rather than deficiencies in the Knowledge Tracing algorithm itself.

3.3 Ethical Considerations

Respondent participation is voluntary and preceded by informed consent. All collected data and test responses are anonymized prior to analysis and used solely for the purposes of this study.

4 Results and Discussions

This chapter presents the analysis and interpretation of the data gathered throughout the study, organized according to the five objectives presented in Chapter 1, and provides

4.1 Learning Challenges and Skills Decomposition for Automata Theory and Formal Challenges

4.1.1 Identified Learning Challenges

4.1.2 Skill Decomposition of Deterministic Finite Automata (DFA)

Based on the review of related

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